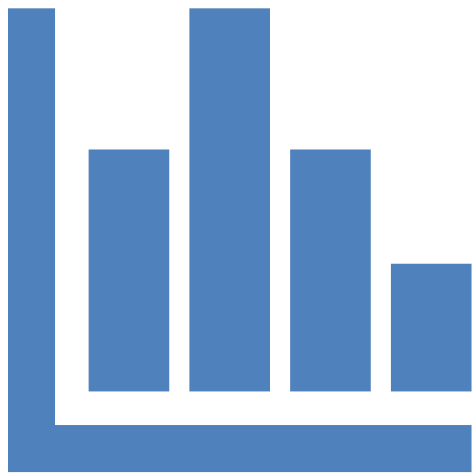




Introduction to Business Analytics

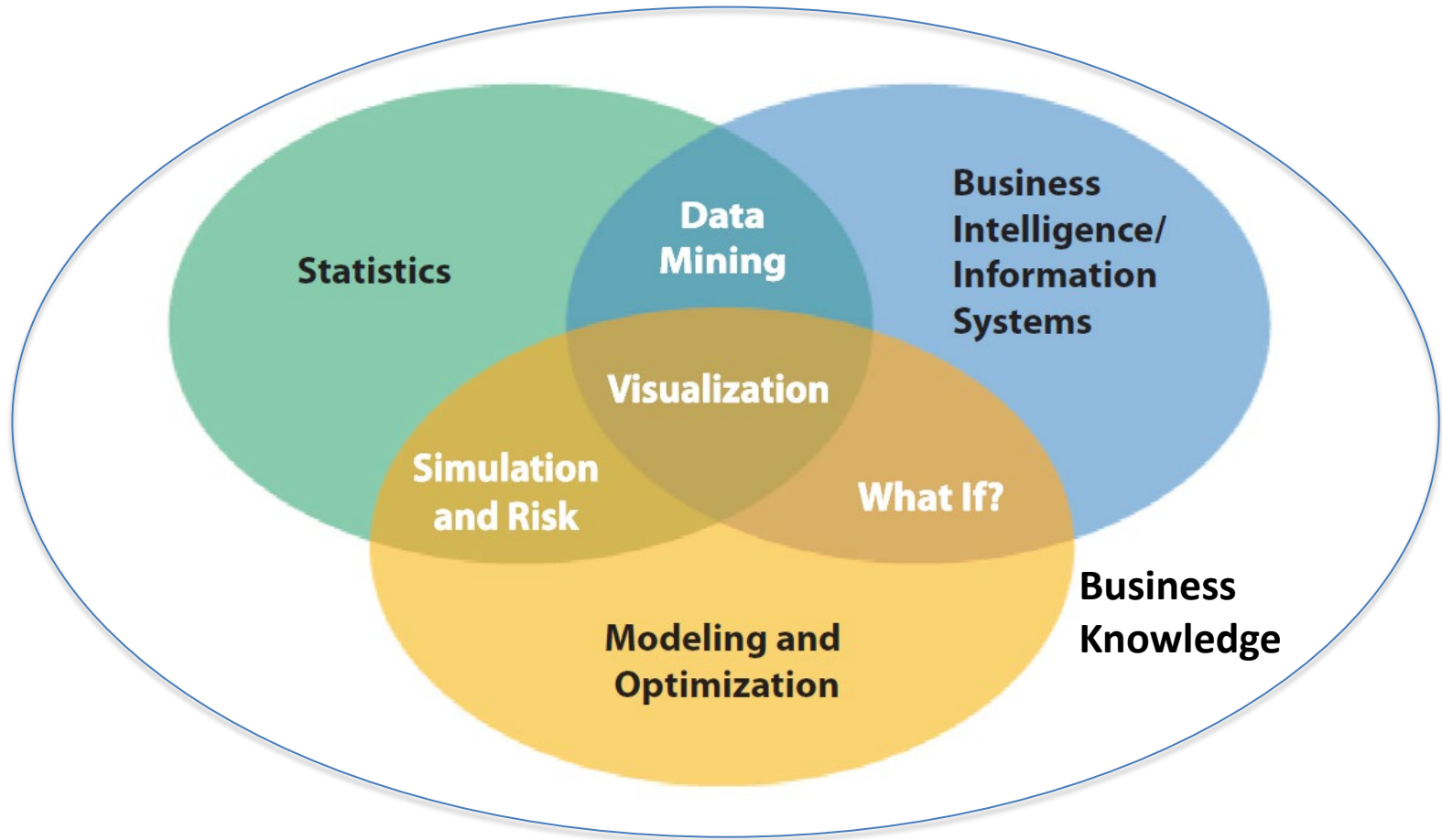
What is Business Analytics?

- **Business Analytics is the use of:**
- **Data:** Collection, processing, and analysis of structured and unstructured data to derive meaningful insights.
- **Information Technology:** Advanced tools and systems to efficiently store, manage, and analyze vast amounts of data.
- **Statistical Analysis:** Application of statistical methods to uncover trends, correlations, and patterns within data.
- **Quantitative Methods:** Mathematical techniques such as optimization, simulation, and linear programming for better decision-making.
- **Mathematical or Computer-Based Models:** Models that simulate real-world scenarios, helping to predict outcomes and guide strategic decisions.

Examples of Business Analytic Applications

Financial Analysis	•Identifying profitable investment opportunities and managing risks.
Pricing	•Optimizing product prices based on customer behavior and market trends.
Marketing	•Predicting customer preferences to create personalized marketing campaigns.
Production	•Streamlining manufacturing processes and improving efficiency with predictive maintenance.
Supply Chain	•Improving inventory management, reducing delays, and optimizing logistics.
Human Resources	•Enhancing employee retention and talent acquisition through data insights.
Sales	•Forecasting sales trends and optimizing sales strategies using customer and market data.
Customer Service	•Improving customer satisfaction by analyzing support data and optimizing response times.
Risk Management	•Using data to assess and mitigate business risks such as fraud and compliance issues.
Healthcare	•Analyzing patient data for better treatment outcomes and operational efficiency.

What does it take to be good at Business Analytics?



Why Business Analytics is Important?

- It is important because:
 - better decision-making
 - Improve efficiency
 - Enhance customer understanding
 - Foster innovation

“It is no longer just an option but a necessity for businesses that want to thrive in the modern economy.”

